

What is left?
What is right?

We and aliens

Imagine: we communicate with aliens sending/receiving signals back and forth

If parity were conserved, every message we send to aliens would remain ambiguous up to mirror reflection. Left and right are not operationally distinguishable.

$$(x, y, z) \leftrightarrow (-x, y, z)$$

Parity violating weak interactions allow
unambiguous definition of left and right!